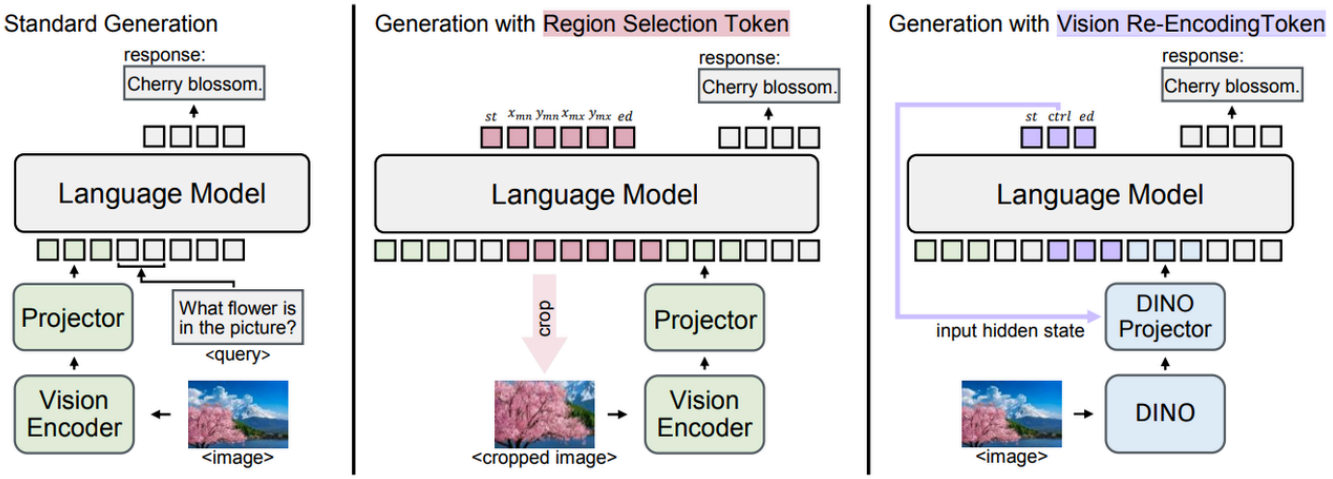
Introducing Visual Perception Token into Multimodal Large Language Model

> #Multimodal #CV #Inference #HybridArchitecture



- What problems does this architecture solve?
  - Currently, the multimodal LLMs lack the ability to adaptively perceive the details of the image.
  - If the model wants the local details of the image, it relies on manual crops.
  - Though this can be done by LLM agents, the efficiency is low.
- How does this architecture work?
  - There are 2 modes of decoding.
  - Region Selection Token
    - Split the image into $k \times k$ patches. This will help the model to locate the patch it wants to see.
    - The context (prefix) will include: encoded compressed full image + NL description.
    - Then, during the decoding, if the model needs to see a patch, it will produce an output with the following order:
      1. Start Token: `<Region_Selection_Start>`
      2. Top left X value.
      3. Top left Y value.
      4. Bottom right X value.
      5. Bottom right Y value.
      6. End Token: `<Region_Selection_End>`
      - All special tokens are added to the vocabulary.
    - After that, using the produced coordinate information, the system will retrieve the patch, feed the patch to visual encoder f_v and projection layer g_v and append to the context.
  - Vision Re-Encoding Token
    - The context (prefix) will include: encoded compressed full image + NL description

        - Then, the model will generate a latent representation with the following order:
            1. Start Token: `<Re-Encode_Start>`.
            2. Latent Representation Token: `<Re-Encoding_Control>` that contains the key information of the context.
                - The latent representation token will be used as K and V, whereas the fine-grained, HD image's latent representation from the encoder will be served as Q. **Cross attention** will be calculated and will be used to extract the useful information from the HD image.
            3. End Token: `<Re-Encode_End>`
        - The NL-based answer will be generated based on the information learned from the prefix context and the cross attention.
        - During training, the `<Re-Encoding_Control>` will not be included in the loss function.
        - To train this and avoid the model cheating (i.e., store nothing in the latent representation but just use the prefix context), masks will be applied to the prefix context of the 50% of the trajectories of the training dataset. Thus, the loss is purely based on the answers that are generated from the latent representation token.
- Experiment Results
    - Outperforming Larger Models: After introducing visual perception tokens, a small model with only 2 billion parameters saw its average performance improve by 30.9% (scores jumped from 0.572 to 0.749), even outperforming the 7-billion-parameter baseline model by 20.0%.
    - Excellent Performance on Specific Tasks: Improvements were particularly pronounced on tasks requiring extremely high visual accuracy. For example, it outperformed the 7B model by 34.6% on spatial reasoning tasks and by 32.7% on fine-grained VQA (Visual Question Answering) tasks.
    - Better results with "free will": Experiments found that the model performed best when it was not forced to use specific tokens, but was instead allowed to autonomously decide whether and which tokens to use based on the current question (Free Choice mechanism).
- Work Cited
    - Paper
    - Chat History
    - NotebookLM